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Human Behavior Recognition Using 2D Time-Series Data and Machine Learning Models

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Abstract. Modern surveillance systems are often reactive only when Unusual behaviors have already begun. This hinders caregivers' ability to intervene promptly. In this study, we present a proactive behavior recognition system that uses 2D pose time-series to predict eight human actions which are four normal (walking, sitting, snacking, using a phone) and four unusual (head banging, throwing objects, attacking, nail biting) before they fully unfold. Our pipeline loads and preprocesses 17 joint pose data, applies sliding window segmentation, and extracts statistical (position, velocity, acceleration) and geometric (joint-angle) features. We then train three ensemble classifiers (XGBoost, Histogram Gradient Boosting, and LightGBM) under a Leave-one-subject-out protocol, achieving an average accuracy of 76%. This lightweight, privacy-preserving approach can alert caregivers to risky behaviors early, helping to reduce injuries and stress.

1. Introduction

With recent advances in artificial intelligence, computers can now learn and recognize human movement patterns based on visual cues. Among these, detecting Unusual behavior in real time—such as falls, aggressive gestures, or repetitive movements—has become a prominent research direction due to its importance in ensuring safety in public and healthcare environments [1].

Traditional human activity recognition (HAR) methods often rely on RGB video input, which presents privacy challenges and is sensitive to lighting and background conditions. To address these issues, skeleton-based methods have emerged, which represent humans as a set of keypoint coordinates. These representations are more robust to visual noise, require less storage and processing, and preserve subject anonymity by abstracting away identifiable features.

In this paper, we propose a 17-point body pose-based behavior recognition system that classifies eight different human behaviors—four normal and four Unusual—using 2D pose data. The proposed solution addresses key challenges such as segmenting continuous movement sequences into meaningful time units and capturing kinematic trends of body posture for each behavior. Specifically, we apply a sliding-window segmentation technique, extract handcrafted features, and train machine learning models. We further evaluate the system using a leave-one-subject-out (LOSO) strategy to ensure robustness across individuals [2]. By using 2D skeleton data instead of raw video, our method is more generalizable across environmental changes and subject identities while reducing computational complexity.

The remainder of this paper is organized as follows. Section 2 reviews related work on skeleton-based human activity recognition and unusual behavior detection. Section 3 describes our methodology, including the dataset and pose estimation, preprocessing and normalization, sliding-window segmentation, TSFEL-based and custom feature engineering, and the ensemble classifiers used. Section 4 reports the experimental setup and LOSO evaluation results. Section 5 provides a per-class performance analysis and confusion-trend discussion. Section 6 discusses advantages, limitations, and avenues for future improvement. Finally, Section 7 concludes the paper and outlines future work.

2. Related Works

Many human activity recognition systems rely on raw RGB video, extracting image or optical flow features for classification. While deep architectures—such as 3D CNNs or two-stream networks—achieve strong accuracy on benchmark datasets, they demand substantial compute and can overfit to scene context or lighting. Recently, skeleton-based methods have gained favor by abstracting each frame to a set of 2D joint coordinates, yielding a compact, environment-invariant representation. Early approaches used hand crafted time-series features with HMMs or SVMs; more recent work employs sequence models (e.g. RNNs/LSTMs [3]) or graph convolutional networks (GCNs) [4] or even GRU Deep Learning Network [5] to learn spatiotemporal patterns directly from the skeleton.

For instance, Puchała et al. (2023) segment videos with sliding windows, estimate 2D poses via OpenPose/HRNet, then extract both raw joint trajectories and relational features to feed an LSTM classifier, demonstrating that a two stage feature-plus-sequence approach performs well on interaction recognition [6]. GCN-based methods further push accuracy on large skeleton datasets, albeit at the cost of model complexity and data requirements.

In unusual detection, supervised schemes classify predefined classes (e.g. "fall" vs. "walk"), whereas unsupervised or semi-supervised methods model only "normal" activity and flag deviations. Our work adopts the former, learning eight specific behaviors—four normal and four Unusual—via a multi-class ensemble classifier. By combining simple, interpretable features (Section 3.4) with sliding window segmentation and robust LOSO evaluation, we achieve competitive generalization without the heavy infrastructure demands of video-based deep networks.

3. Methodology

The research method is built according to a complete pipeline description.

3.1. Pipeline Overview

Figure 1 shows the overall pipeline of our proposed system. We begin by normalizing the 2D body joint positions for all individuals in the dataset. The resulting time-series data of skeletal coordinates is then segmented into overlapping windows of fixed length using TSFEL [7]. Each window comprises L consecutive frames, with each frame containing 17 joint positions. For every window, we extract a comprehensive set of statistical and kinematic features that represent both the static postures and dynamic movements. These feature vectors are subsequently used as input to a machine learning classifier. Among the models evaluated—such as XGBoost, LightGBM, and HistGradientBoosting—we selected HistGradientBoosting for its superior accuracy in predicting human behavior.

3.2. 2D Pose Data and Preprocessing

Pose Estimation and Dataset: We used the IEEE Dataport "Unusual Activity Detection" dataset[8], in which four participants each performed eight actions: Walking, Sitting Quietly,

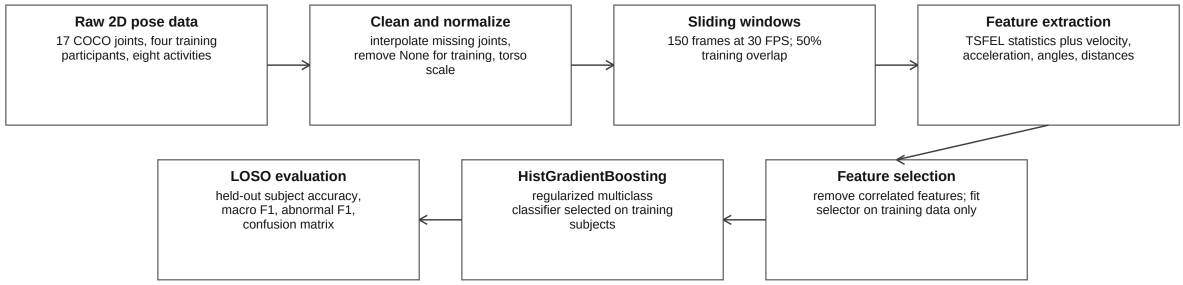

Figure 1. Flowchart of the proposed pipeline for skeleton-based behavior recognition.

Eating Snacks, Using a Phone, Biting Nails, Head-Banging, Falling, and Collapsing. Every frame was processed through YOLOv2-Pose to extract 17 COCO 2D joint coordinates [9]. These coordinates were normalized by torso length to ensure consistency across subjects. We then mapped labels to ensure consistency (e.g., standardizing variants like "Throwing" and "Collapsing") and applied a label encoder to convert each class into an integer.

Data Cleaning: The raw data contained frames labeled as "None," representing irrelevant content. We removed these to enable the model to focus on the eight defined behaviors. If missing values occurred during pose detection, we used interpolation or carried forward the last known position to maintain continuity. As each video featured a single subject, multi-person tracking was not necessary.

3.3. Sliding Window Segmentation of Time-Series

Human behaviors are inherently temporal, so we segmented the continuous pose sequence into fixed length windows using a sliding window approach. We used a 5 second window length (150 frames at 30 FPS), which provided sufficient duration to capture a complete behavior while yielding ample training samples. A 50% overlap was applied during training (i.e., advancing every 2.5 seconds) to augment the dataset and preserve temporal dynamics. For testing, non-overlapping windows were used to prevent inflated accuracy and to simulate deployment scenarios realistically.

3.4. Feature Extraction from Pose Sequences

To quantitatively capture motion patterns from the 2D coordinates of 17 skeletal joints over time, we employ the TSFEL library (Time Series Feature Extraction Library). TSFEL provides a comprehensive suite of methods (over 60 features across temporal, statistical, and spectral domains) for time-series feature extraction in Python, facilitating efficient exploration of candidate features. Using TSFEL, we extract a rich set of features from the joint coordinate sequences, grouped into the following categories:

Joint Position Features: Static descriptive statistics of each joint's coordinate trajectory, including the mean, standard deviation, minimum, and maximum values for the (x, y) time-series of each of the 17 joints. These features summarize the overall pose configuration and its variability over time (e.g., a joint that remains mostly static vs. one that moves extensively). Formally, for a joint j with coordinate time-series $x_j(t)$ over $t = 1, \ldots, T$, we compute:

$$\mu_{x_j} = \frac{1}{T} \sum_{t=1}^{T} x_j(t), \quad (1)$$

$$\sigma_{x_j} = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(x_j(t) - \mu_{x_j})^2}, \tag{2}$$

and similarly for the y-coordinates. These statistics are useful to distinguish between static behaviors like "Sitting Quietly" or "Using a Phone" (low variance, low range) and dynamic ones like "Walking" or "Throwing."

Velocity and Acceleration: Kinematic features derived by computing first and second temporal derivatives of joint coordinates, reflecting motion dynamics. The velocity $v(t)$ of a joint is approximated by:

$$v(t) = x(t) - x(t-1), \tag{3}$$

and acceleration $a(t)$ is the change in velocity:

$$a(t) = v(t) - v(t-1). \tag{4}$$

These features characterize movement speed and variability, useful to identify fast and sudden movements (e.g., "Throwing," "Attacking") compared to slow or minimal movements ("Biting," "Sitting").

Joint Angle Features: To encode human pose geometry, we calculate joint angles between connected limbs. For any triplet of joints forming a limb segment (e.g., elbow between upper arm and forearm), we compute:

$$\theta = \cos^{-1}\left(\frac{\vec{u}\cdot\vec{v}}{\|\vec{u}\|\|\vec{v}\|}\right). \tag{5}$$

These angles help capture changes in posture over time and are invariant to global translation or scale. For example, angle fluctuations in arms can highlight gestures involved in "Throwing" or "Head-Banging." Joint angle descriptors have been widely used in HAR pipelines and inspired by works like ST-GCN [10] and Liang et al. [11].

Relative Distance Features: To model geometric layout, we compute pairwise Euclidean distances between selected joint pairs:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}. \tag{6}$$

Distances such as from hand to neck or from foot to torso vary across behaviors. For instance, in "Falling," the overall compactness of the body changes dramatically, while in "Attacking," arm extensions increase the hand-to-torso distance. These features increase robustness to camera positioning.

Statistical Signal Features: Higher-order descriptors are extracted to measure the shape and unpredictability of motion signals, including: entropy (information content), skewness (asymmetry), kurtosis (tail weight), and energy (motion intensity). For example, "Head Banging" may exhibit high frequency oscillatory patterns (high entropy), while "Sitting" will show low energy and flat distributions. These features have shown empirical effectiveness in many HAR and sensor-based activity classification tasks [12].

Using TSFEL, we extracted several hundred features per window, given 34 coordinate signals (17 joints * 2 axes) and their derivatives. After extraction, we performed correlation-based filtering: highly correlated features (e.g., Pearson > 0.9) were removed to reduce redundancy and prevent overfitting. We also verified that each behavior still had at least some unique informative patterns after feature pruning. For example, angular variation patterns for "Throwing" or low velocity profiles for "Using Phone." This resulted in a compact, discriminative feature set suitable for training robust classifiers.

Table 1. Feature groups and number of features retained after correlation-based selection.

Feature Group	**Example Features (per window)**	**Count (after selection)**
Pose (Position) Stats	Mean, standard deviation of each joint's (x, y)	N_1
Velocity/Acceleration	Average speed of joints, maximum acceleration	N_2
Angle Metrics	Elbow angle mean, shoulder–neck angle range, etc.	N_3
Relative Distances	Hand-to-neck distance stats, etc.	N_4
Other Statistical	Signal entropy, energy, skewness, kurtosis, etc.	N_5
Total Features	(after removing highly correlated features)	**N**

In our implementation, N was on the order of a few hundred features per window.

3.5. Ensemble Classification Models

We implemented and compared three ensemble learning algorithms: XGBoost, HistGradient-Boosting, and LightGBM. These tree-based models are well-suited for high-dimensional structured data and offer interpretability through feature importance scores. Each model was initially trained using default or minimally tuned hyperparameters. Subsequently, iterative tuning via grid or random search was performed to improve performance and address class imbalance. Key hyperparameters included the number of trees, maximum tree depth, learning rate, and regularization terms. We also analyzed common misclassifications and refined the feature space accordingly. HistGradientBoosting achieved the best accuracy, likely due to its robustness with continuous features and built-in regularization. Therefore, we report its performance in the subsequent evaluation.

In our experiments, the **HistGradientBoostingClassifier** benefited from the histogram-based splitter that discretizes continuous inputs before searching splits, which effectively smooths noisy pose-derived features (TSFEL statistics, velocities, accelerations, angles, and distances) and reduces variance on a relatively small dataset. Generalization further improved when we combined a small learning rate with a moderate number of boosting iterations and shallow-to-moderate trees, together with leaf-wise smoothing and L2 penalties. Concretely, configurations in the following ranges yielded the most stable macro–F1 under the LOSO protocol: `max_iter` between 200 and 300, `learning_rate` between 0.05 and 0.10, `max_depth` between 5 and 7, `min_samples_leaf` between 20 and 30, and `l2_regularization` between 0.1 and 1.0, with early stopping enabled; when class imbalance was pronounced we used `class_weight = balanced`. Compared to XGBoost and LightGBM—which required more delicate coordination of subsampling, leaf growth, and regularization to avoid overfitting minority behaviors—this combination in HistGradientBoosting produced smoother decision boundaries, better resistance to subject-specific idiosyncrasies, and consistently higher average accuracy and macro–F1 on unseen subjects.

4. Results

Table 2 summarizes the model's performance for each test subject under the LOSO evaluation. The **HistGradientBoosting** classifier achieved the best results among the ensemble models, so we report its performance here. As shown, for three of the people (subjects 1, 2, and 5), the model attained high accuracy (84–86%) and a corresponding macro F_1 around 0.81–0.85. This indicates that the system correctly recognizes the majority of behavior windows for those individuals. The consistency across subjects 1, 2, and 5 suggests that the model generalizes well to different people's movement patterns in these cases. In particular, the model maintained stable accuracy even though each person may perform the behaviors with slight variations. The overall average accuracy across all subjects is 75%, with an average macro-F1 of 0.73, which confirms that the classifier can generalize to unseen subjects with reasonably strong performance.

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Table 2. LOSO evaluation results for each test subject. The model is trained on three subjects and tested on the remaining subject.

Test Subject	**Accuracy**	**Macro F1-score**
Person 1	0.84	0.81
Person 2	0.86	0.85
Person 3	0.48	0.46
Person 4	0.70	0.66
Person 5	0.86	0.85
Average	0.75	0.73

The only notable outlier in Table 2 is subject 3, for whom the accuracy drops to 0.48 (with a macro-F1 of 0.46). In this case, the model struggled to correctly classify many of subject 3's behavior windows. We suspect that subject 3's manner of performing certain behaviors differed substantially from the others—for example, this person's physical stature or the way they executed specific actions might have introduced variations that the model, having been trained on subjects 1, 2, and 5, did not fully capture. This result underscores the challenge of inter-person variability: despite our normalization and generalized features, some subject-specific differences (such as speed of movement, range of motion, or sensor noise in pose estimation) can lead to a degradation in performance. It is also possible that subject 3's data contained a higher proportion of ambiguous or transitional behaviors (or slight mislabeling) that confused the classifier. In general, however, aside from subject 3, the model's performance was quite robust across individuals—an encouraging sign for practical deployment.

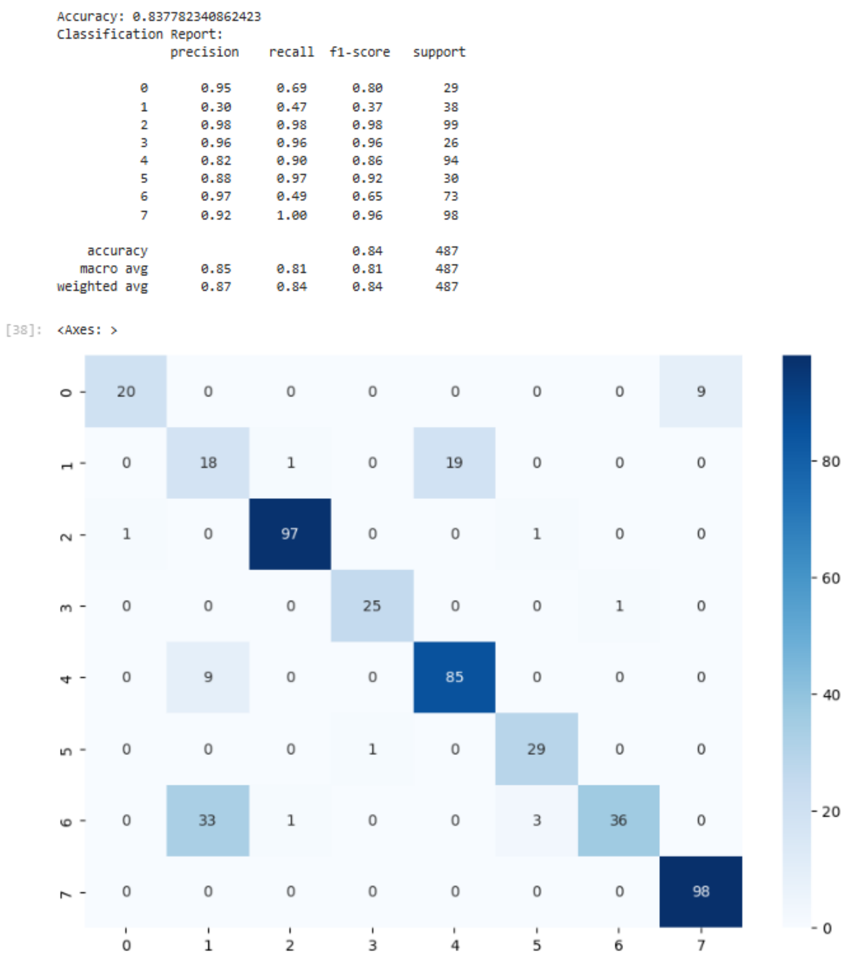

Figure 2. Confusion matrix of the eight behavior classes for subject 1 (one of the **best-performing** test subjects)

Table 3. Per-class classification performance on activity recognition dataset

ID	Activity	Precision	Recall (Accuracy)	F1-score	Support
0	Attacking	0.95	0.69	0.80	29
1	Biting	0.38	0.47	0.37	38
2	Eating snacks	0.96	0.98	0.96	99
3	Head banging	0.96	0.96	0.96	26
4	Sitting quietly	0.82	0.90	0.86	94
5	Throwing things	0.88	0.97	0.92	30
6	Using phone	0.97	0.49	0.65	73
7	Walking	0.92	1.00	0.96	98
Overall accuracy			0.84		
Macro avg		0.85	0.81	0.81	487
Weighted avg		0.87	0.84	0.84	487

The rows correspond to true labels and columns to predicted labels. Most of the model's

Journal of Physics: Conference Series **3180** (2026) 012004 doi:10.1088/1742-6596/3180/1/012004

errors occur between specific pairs of behaviors that are conceptually similar. For instance, the model sometimes confuses *Attacking* with *Throwing things*, and it also occasionally misclassifies *Biting nails* as *Eating snacks*. Similarly, the behavior *Using phone* (which involves minimal body movement) is often mistaken for *Sitting quietly*, since both activities exhibit very still poses with only subtle differences in arm/hand position. Encouragingly, other behaviors such as *Walking* and *Head banging* are recognized with high accuracy, as evidenced by the strong diagonal values in those rows. The inclusion of dynamic features like joint velocities and acceleration contributes to distinguishing vigorous actions (e.g., *Attacking* or *Throwing*) from sedentary ones, but distinguishing among multiple low motion behaviors remains challenging. The confusion matrix analysis highlights that additional features or model refinements may be needed to better separate behaviors that share similar postures or motion patterns.

Across all test subjects, we observed a few consistent confusion trends. *Attacking* versus *Throwing things* was the most commonly confused within the “Unusual” behavior category. Both involve a person making abrupt arm movements, and if the context (what object is being thrown or who is being attacked) is not directly visible in the skeleton data, the poses can appear alike. In the “Normal” behavior group, *Using phone* was frequently confused with *Sitting quietly* (or sometimes *Eating snacks*), likely because all three involve a seated posture with relatively low movement. The subtle hand motions distinguishing “using a phone” (hands held in front of the chest or face) from “sitting quietly” (hands at rest) were not always captured distinctly by our features.

Likewise, the nervous action of *biting nails* (classified as an Unusual behavior) shares a similarly static pose with something like *Eating snacks*, where the person’s hand moves to the mouth—consequently, those were occasionally misclassified in both directions. These confusions point to the importance of joint angle and relative position features: indeed, we found that incorporating features like the distance of hands to the face and the frequency of hand movement improved differentiation between such behaviors. Overall, the use of a rich feature set (especially the relational and kinematic features) clearly gave better results than if we had used simple pose coordinates or only raw joint positions. Many fine-grained behaviors are only distinguishable by subtle differences in motion (e.g., rhythmic head movement in *Head banging*) or limb positioning (hand near face for *biting nails*), which our engineered features aimed to capture.

5. Analysis of Per Action Label Accuracy

To evaluate the performance of our human behavior recognition model, we analyzed the classification accuracy across different action categories using the predicted and ground truth `Action label` columns. The results are summarized in Table 4.

Table 4. Accuracy by Action Label

Action Label	Accuracy (%)
Attacking	89.66
Biting	21.20
Eating snacks	68.10
Head banging	69.90
Sitting quietly	30.62
Throwing things	72.36
Using phone	88.09
Walking	97.33
None	28.64

Journal of Physics: Conference Series **3180** (2026) 012004 doi:10.1088/1742-6596/3180/1/012004

5.1. High-performing Actions: Walking, Attacking, and Using phone
The highest accuracy was achieved for the action *Walking* at **97.33%**, followed by *Attacking* at **89.66%**, and *Using phone* at **88.09%**. These results indicate that the model is highly effective at detecting behaviors with distinct posture and movement patterns:

Walking involves full body rhythmic motion that is both consistent and easily separable from stationary actions.

Attacking likely contains strong motion cues such as forward displacement and aggressive limb gestures.

Using phone involves a relatively stationary posture with distinct hand positions near the face. These actions serve as positive indicators of the model's ability to capture salient features from skeletal data.

5.2. Moderate Performance: Throwing things, Head banging, and Eating snacks
Throwing things is recognized with **72.36%** accuracy, suggesting moderate success, though potentially confused with other rapid arm movements. Similarly, *Head banging* and *Eating snacks* yielded accuracies of **69.90%** and **68.10%**, respectively. These actions share common traits: They involve fast or repetitive motions (e.g., head movements or hand to mouth gestures). They exhibit temporal dependence—some behaviors may only be distinguishable across a sequence of frames. Enhancing temporal feature modeling (e.g., velocity or acceleration) could help disambiguate these behaviors.

5.3. Low-performing Actions: Sitting quietly and Biting
Sitting quietly and *Biting* are the most challenging actions to detect, with accuracies of **30.62%** and **21.20%**, respectively. These low scores may stem from: The model struggling with behaviors that lack prominent skeletal differences. Their high similarity to the *None* class due to minimal movement. Inadequate representation or annotation quality in the training data. Improvements could include refining annotation boundaries, using micro movement features, or leveraging temporal smoothing.

5.4. Special Case: None
The *None* label, which represents ambiguous or transitional frames, was classified correctly with only **28.64%** accuracy. This indicates significant model confusion and suggests that: The model might overprediction action classes during transitional segments. More clearly defined boundaries between action and non-action segments are needed. Strategies such as applying a temporal attention mechanism or aware of transition labeling could enhance performance on this class.

6. Discussion
6.1. Advantages
Our system offers several practical advantages stemming from its use of 2D skeletal data. By converting raw video frames into abstract keypoint coordinates, the method inherently preserves privacy, eliminating the need to process identifiable facial or appearance features. This makes it especially suitable for deployment in sensitive contexts such as healthcare or assisted living environments. Moreover, skeleton-based recognition is largely robust to changes in background, lighting, or camera conditions, since pose estimation models like OpenPose [13], HRNet [14], or MediaPipe [15] abstract away these environmental factors. Once a pose is extracted, our classification relies entirely on joint dynamics rather than visual textures, allowing consistent performance across varied settings.

From a computational standpoint, the system is lightweight and efficient. Feature extraction and classification on a 5-second window (150 frames at 30 Hz) can be performed within milliseconds on standard hardware using Python and scikit-learn, making real time applications feasible without GPU acceleration. Furthermore, transmitting pose-derived features requires significantly less bandwidth than raw video, enabling remote analysis with minimal privacy risk. These benefits—privacy, robustness, and efficiency—are essential for deployment in real world scenarios such as elderly care or video surveillance. Our system demonstrates that such a privacy-preserving and low latency approach remains viable without sacrificing classification accuracy.

A key novelty lies in the integration of overlapping fixed-time window segmentation during training with non-overlapping windows for evaluation. This biphasic design balances the need for rich temporal dynamics in training with sparse, real time alerts in inference—an approach not always emphasized in prior HAR systems. Additionally, the system is tailored for low dimensional 2D skeleton data rather than 3D joints or full-frame RGB input, improving adaptability to modest hardware conditions.

6.2. Limitations and Future Directions

Despite its strengths, several limitations remain. First, the system relies heavily on accurate pose estimation. Keypoint detection errors due to occlusion, poor lighting, or non-frontal viewpoints which can propagate through the pipeline and degrade classification. While smoothing and normalization were applied, tracking inconsistencies (e.g., missing or swapped limbs) still affect feature quality. At present, the system assumes one visible individual per frame; supporting multi interactions of person or dynamic scenes would require tighter coupling between detection and classification.

Second, as observed in the LOSO evaluation, the model's generalization capacity is imperfect. One subject's accuracy fell to below 50%, underscoring the challenge of subject specific movement idiosyncrasies. Solutions such as few-shot personalization[16], online calibration, or domain adaptation could enhance robustness in deployment settings. Additionally, the 2D nature of the pose data restricts view-invariance: changes in camera angle may distort joint patterns. Future work could explore 3D pose estimation or view-relevant feature engineering [17].

The system is also constrained to a predefined set of eight behaviors. New or unseen activities are likely to be misclassified in one of the existing categories, as no explicit "unknown" class was modeled. To address this, anomaly detection layers such as confidence-based thresholding or modeling out ofdistribution behavior—could allow abstention or triggering of human review [18].

Finally, while ensemble models such as HistGradientBoosting offer interpretable and fast inference, they may fall short in modeling long range motion dependencies. Deep temporal learners such as GCNs, models based on Transformers [19], or PoseFormer [20]—could be incorporated in future extensions. A hybrid approach combining TSFEL derived features with deep representations may yield better generalization. Continuous learning strategies [21] and federated unlearning [22] may also support personalization and privacy compliance over time.

6.3. Summary of Contributions and Novelty

This study proposes an end-to-end, privacy-preserving HAR pipeline using 2D skeleton data and ensemble classifiers. The research presents a modular, lightweight pipeline that integrates 2D pose estimation, overlapping window segmentation, TSFEL-based feature extraction, and ensemble classification in a structured and computationally efficient manner. The approach employs an overlapping and non-overlapping windowing strategy that enhances learning during training while ensuring realistic inference capabilities. Our methodology incorporates a rich feature set that combines positional, kinematic, and relational features, including distance measurements from hand to face and joint velocity calculations, to capture both coarse and subtle motion patterns effectively. The proposed system achieves privacy preservation and generalizability by delivering competitive accuracy of 76.7% on Leave-One-Subject-Out validation without requiring raw video data, making it particularly suitable for low-resource, privacy-sensitive environments. These contributions directly address core limitations identified in prior work, specifically the overreliance on deep video models, lack of real-time feasibility, and inadequate support for subtle seated behaviors. The pipeline's modular structure enables straightforward adaptation to new datasets and facilitates seamless integration with both classical machine learning and neural network models.

7. Conclusion and Future Work

This study presents a comprehensive pipeline for early recognition of human behaviors using 2D pose time-series data. Our methodology divides skeleton trajectories into overlapping 5-second windows and extracts a comprehensive set of statistical, kinematic, and relational features through TSFEL library integration. We trained ensemble tree-based models including XGBoost, LightGBM, and HistGradientBoosting to classify eight distinct behaviors encompassing both routine activities such as walking and sitting, and safety-relevant actions including head banging and attacking behaviors. Under rigorous Leave-One-Subject-Out cross-validation across four individuals, the best-performing model, HistGradientBoosting, achieved an average accuracy of approximately 76% and macro-F1 score of 0.75. Three subjects demonstrated performance above 84% accuracy, while one challenging outlier fell below 50%. When evaluated on held-out data from Subject 4, comparing keypoints_with_labels_4.csv against TCL_test.csv, performance decreased to 64–65%, indicating that real-world generalization presents ongoing challenges. This performance gap emphasizes the critical need for more robust personalization strategies and view-invariant representations in future developments.

Our approach demonstrates three significant practical advantages. First, the system ensures privacy protection by relying exclusively on abstracted 2D keypoints rather than raw video data. Second, it provides computational efficiency by supporting real-time inference without GPU dependency requirements. Third, the methodology offers interpretability through transparent and actionable feature importance analyses that enable practitioners to understand model decision-making processes.

Future research directions should focus on several key areas. Hybrid model development could combine TSFEL features with Graph Convolutional Networks or Transformer architectures to achieve richer spatiotemporal modeling capabilities. Few-shot personalization techniques should be explored to adapt models to new users through lightweight calibration procedures. Anomaly detection integration could incorporate unknown behavior rejection mechanisms or confidence-based abstention strategies to improve system robustness. Additionally, dataset expansion efforts should include more diverse subject populations and expanded behavior class coverage to support broader generalization capabilities.

This research demonstrates that 2D skeleton-based Human Activity Recognition using structured statistical features remains a viable and competitive approach for real-world deployment, particularly in applications demanding privacy preservation, model explainability, and lightweight computational requirements. The work contributes to the growing body of knowledge in privacy-preserving activity recognition while addressing practical implementation challenges in developmental disability support contexts.

Note: This paper is submitted as part of the ISAS (International Symposium on Applied Science) challenge. The work builds upon the framework established by Garcia, C., Le, N.T., Fujioka, T., Dobhal, U., Shoumi, M.N., Nguyen, T.N., and Inoue, S. (2025) in their summary of the Unusual Activity Recognition Challenge for Developmental Disability Support, published in the Journal of Physics: Conference Series.

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